Bingxi (Paul) Jiang

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EDUCATION

University of California, Irvine Expected June 2027
Bachelor of Science, Computer Science | GPA: 3.835/4.0 Irvine, CA
Relevant Coursework: Artificial Intelligence, Machine Learning, Information Retrieval, Probability and Statistics, Data Structures & Algorithms

TECHNICAL SKILLS

Languages Python, C++, SQL, JavaScript, TypeScript, Java, Go, Bash
ML & Deep Learning PyTorch, TensorFlow, scikit-learn, XGBoost, NumPy, Pandas, Transformers, Hugging Face, NLP, Computer Vision
Generative AI & LLMs LLMs, RAG, Embeddings, Vector Search, Fine-Tuning, LoRA, SFT, RLHF, Agents, MCP, Quantization, LLM Evaluation
Backend & Systems Node.js, FastAPI, REST APIs, gRPC, WebSockets, Microservices, Distributed Systems, Async Processing, Redis, PostgreSQL, Linux
MLOps & Data Spark, PySpark, Kafka, Airflow, Docker, Kubernetes, MLflow, AWS, GCP, ONNX, vLLM, CUDA, CI/CD, Prometheus, Grafana

PROFESSIONAL EXPERIENCE

AI Engineer Intern June 2026 - September 2026
Dassault Systèmes San Diego, CA

- Built a closed-loop LLM workflow-migration system with Playwright MCP and structured validation, cutting migration time by 69%.
- Benchmarked Groovy, manual coding, OSS LLMs, and IDE agents; closed-source agents delivered the best speed, reliability, and output quality.
- Added schema validation, retry logic, execution feedback, failure analysis, and prompt refinement to automatically repair generated workflows.
- Built reusable TypeScript modules for nested iFrames, grids, sessions, and APIs, reducing new workflow implementation from ~1 week to ~1 day.
- Cut a 23-step enterprise workflow from 15–20 min to 3–4 min while expanding validation checkpoints from 6 to 23.

Research Assistant January 2026 – Present
Relational Cognition Lab Irvine, CA

- Built a PyTorch evaluation pipeline for 100K+ prompts across Gemma, Qwen, and Llama, standardizing large-scale model benchmarking.
- Scaled inference with batching, caching, multi-GPU execution, and distributed jobs to support reproducible model experimentation.
- Evaluated models with accuracy/F1, correlation, ranking, and probe metrics using confidence intervals, significance tests, and effect sizes.
- Probed transformer representations with classifiers, cosine similarity, logit lens, and activation patching, revealing relational signals.

Software Engineer Intern April 2026 - June 2026
GTechFin Inc (B2C Fintech startup) New York, NY

- Built an end-to-end LLM content pipeline spanning retrieval, generation, validation, and rendering, cutting production time from ~3h to ~20m.
- Implemented a custom cosine-similarity vector store after benchmarking Recall@K, Precision@K, LLM-as-judge, cost, and retrieval control.
- Replaced paid vector-search infrastructure with an in-house retrieval layer, reducing incremental vector-search infrastructure cost to near zero.
- Added schema validation, quality scoring, retries, and API fallbacks, improving generation success, output validity, and quality by up to ~10%.

Software Engineer Intern September 2025 - December 2025
Fresh Road Inc (B2B SaaS startup) San Jose, CA

- Built a real-time production LLM platform with Node.js, WebSockets, Twilio, Deepgram, OpenAI, and CRM integrations across 25 deployments.
- Benchmarked LLM providers on quality, latency, and cost with A/B testing, helping cut inference cost 32% and p95 latency 18%.
- Applied response caching, prompt compression, and INT4 quantization to optimize inference efficiency under real-time latency constraints.
- Orchestrated transcription, summarization, CRM sync, and LLM jobs with BullMQ/Redis, reducing pipeline failure rate from 10% to 3%.
- Monitored response quality, latency, cost, and failures through dashboards and Sentry, accelerating production issue detection and diagnosis.

Computer Vision Algorithm Engineer Intern June 2025 - September 2025
Ping An Technology Guangdong, China

- Built an end-to-end multimodal damage-recognition pipeline from 2.27M-image data curation to Qwen2.5-VL training and production inference.
- Cleaned, deduplicated, augmented, balanced, and QA'd training data with hard-example mining to improve dataset quality and robustness.
- Fine-tuned Qwen2.5-VL with LoRA using multi-GPU DDP, mixed precision, and hyperparameter tuning, lifting recall 10% and precision 6%.
- Optimized PyTorch/ONNX GPU inference with INT8 quantization and dynamic batching, cutting p95 latency 38% and raising throughput 50%.
- Deployed Dockerized, Redis-cached inference on Kubernetes for 10K+ daily requests with quality, error, and model-drift monitoring.

Research Intern January 2025 – June 2025
DeepEM Lab Irvine, CA

- Built an ML pipeline on ~210K battery records to predict cycle-life qualification, improving test accuracy from a 52% baseline to 87%.
- Benchmarked logistic regression, XGBoost, neural networks, and SVM variants to select the best-performing model family.
- Standardized preprocessing, cross-validation, hyperparameter tuning, and MLflow tracking for reproducible model selection and experiments.

PROJECTS

Longevity LLM Benchmark: Domain LLM Evaluation Suite github.com/Bingxi-Jiang/LongevityLLM-Benchmark

- Designed a 7-task evaluation suite across classification, A/B test, set generation, and regression, scored with accuracy, mean set F1, MAE.
- Built ontology-grounded reasoning scorer validating gene, allele, MP-ID, and longevity-pathway logic beyond final-answer accuracy.
- Built provider-agnostic evaluation harness for GPT, Claude, Gemini, and fine-tuned Qwen with cached inference, and token/latency metrics.

Noted: Real-Time Multimodal Transcription & Memory System github.com/Bingxi-Jiang/Noted

- Architected real-time transcription pipeline streaming 16kHz PCM audio via WebSocket to Deepgram Nova-2, achieving <300ms latency.
- Engineered multimodal memory system combining transcripts, speaker diarization, and Gemini Vision analysis for context preservation.
- Implemented cross-session RAG with embedding similarity search over transcript chunks, enabling context-aware retrieval across folders.